

Partner-Specific Affective Precision in Social Active Inference

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Abstract. Social interaction requires agents to predict what others will do, but multi-agent settings add a second problem: deciding how confidently each partner model should guide action. We propose that affect can serve this role. Using active inference, we extend the affect-as-expected-precision account of Hesp et al. [10] from a single-agent setting to repeated interactions with multiple social partners. A focal agent plays a multi-partner trust game while maintaining separate beliefs about each of four partners and a partner-specific affective precision estimate updated from prediction error in that partner’s behavior. This estimate modulates the precision of policies involving that partner, making the agent more or less decisive when acting on a given partner model. Simulations show that partner-specific affective precision functions as an action-confidence signal for social inference. It tracks predictive reliability more strongly than realized payoff, improves payoff and reduces policy entropy in graded decision regimes where the policy space is open enough for precision to matter, and shifts engagement toward more predictable partners in partner-choice settings. A tracked-only lesion—which preserves partner-belief accuracy while removing precision modulation—shows that these effects arise from action selection rather than improved belief quality. The mechanism has a volatility limit: after abrupt behavioral change by a partner, precision sharpens action around stale partner beliefs and reduces performance, while gradual change is neutral. Affect, in this account, augments social multi-agent inference by regulating how confidently inferred partner states are expressed in action—but its value depends on whether the social world changes faster than partner beliefs can recalibrate.

Keywords: Active inference · Affect · Social cognition · Multi-agent systems · Trust

1 Introduction

Social interaction is often framed as a theory-of-mind problem: agents must form expectations about what others know, intend, and are likely to do [7, 8]. In multi-agent environments, however, prediction is only part of the challenge. A focal agent may have a well-supported model of one partner, a fragile model of another, and an actively changing model of a third. Adaptive social behavior therefore requires not only accurate partner inference, but also a way to decide

how strongly each inferred partner state should guide action. Being precise about a model you have good grounds for, and cautious about one you do not, is not just statistically sensible—it is behaviorally consequential. Over-committing to an uncertain model causes brittle choices; under-committing to a well-validated one wastes hard-won evidence.

This paper treats affect as a solution to that second problem. Rather than modeling affect as a reward signal or a global mood, we model it as a partner-specific estimate of social model fitness: a running signal of how well the agent’s model of a given partner is supporting prediction and guiding confident action. In this view, affect augments social cognition by regulating the confidence with which inferred partner states are expressed in policy selection. A predictable adversary and an unpredictable ally may deliver similar realized rewards while presenting very different model-reliability situations; the affective signal, on this account, tracks the reliability difference rather than the reward difference.

We use active inference as the formal machinery because it naturally separates beliefs about hidden social states from the precision with which policy beliefs are expressed in action [4, 5, 13]. Agents infer hidden states by minimizing variational free energy, evaluate candidate policies by their expected free energy [6, 3], and select actions through a precision-weighted policy posterior. Policy precision γ is the parameter that controls how decisively the agent commits to its most-preferred policy: high precision concentrates action; low precision hedges across alternatives.

Hesp et al. [10] provide a formal route from affective valence to inferred model fitness, treating affect as an estimate of how well the agent’s generative model is supporting action. Rather than treating γ as a fixed hyperparameter, they model the agent as actively inferring a higher-order belief β over its expected action precision—the subjective fitness of its current generative model. An affective charge signal ϕ drives updates to β : it rises when the model performs better than a prior baseline and falls when it performs worse. This makes affect a derivative of model fitness—a signal that responds to improvements and degradations in predictive performance rather than to reward history—which is what allows it to be positive toward a reliably hostile partner while remaining informative about when to act confidently.

Our extension is to factorize this affective precision signal over social partners. A focal agent interacting with multiple partners maintains a separate belief state and a separate β_k estimate for each partner k , both updated from that partner’s behavioral evidence alone. This allows the agent to modulate action confidence differently across its social portfolio: more decisive toward partners its model predicts well, more cautious toward those whose behavior remains uncertain. We test this in a repeated trust game [2, 11] in which four partners vary in type and disposition across binary and graded decision regimes, partner-choice conditions, and betrayal scenarios. The trust game is a natural testbed because partner types are hidden and must be inferred—so there is a genuine gap between social knowledge and actionable confidence—and because varying the policy space lets

Table 1: Focal agent payoffs in the binary trust game.

	Partner cooperates	Partner defects
Agent cooperates	3	-1
Agent defects	5	1

us test whether the affective precision channel operates where the theory says it should.

Across all conditions, we ask one underlying question: does partner-local affective precision function as a useful action-confidence augmentation for social inference, and when does it break down? Results show that the affective precision channel tracks partner predictability rather than partner value, produces behavioral improvements in decision-open regimes specifically through action sharpening rather than better beliefs, and reorganizes partner engagement around reliability before aggregate payoff separates. The mechanism fails—predictably and informatively—when a partner changes abruptly, because affective precision sharpens action before partner beliefs have recalibrated, making the agent decisive at the wrong time.

2 Methods

2.1 Trust game

A focal active-inference agent plays a repeated trust game [2, 11] with four partners over an episode of up to 200 rounds. Each partner has a latent *type* (cooperator, reciprocator, exploiter, or random) and a latent *disposition* toward the agent (trusting, neutral, or hostile)—neither directly observable; both must be inferred from behavioral evidence.

In the binary game both players choose to cooperate or defect. Table 1 gives the focal agent’s payoffs. The structure creates a genuine inference problem: defecting is individually tempting regardless of partner behavior, but a cooperative partner is worth engaging, so the agent must infer the partner’s type and disposition to choose well. In the graded regime, the binary choice is replaced by a discretized investment level; the agent commits a resource along a continuous scale rather than making a binary commitment. This distributes policy probability mass across several options, giving the affective precision channel meaningful room to influence which level is selected. A partner-choice regime adds a prior decision each round: the agent selects which partner to engage before choosing an action, turning approach and avoidance into explicit policy outputs.

2.2 Per-partner generative models

The focal agent maintains a separate discrete generative model $\mathcal{M}_k$ for each partner k , rather than a single pooled social model. This keeps evidence about

each partner’s behavior independent and—critical for the analysis below—allows the confidence with which each model is acted upon to differ across partners. For partner k :

$$s_k = (s_k^{\text{type}}, s_k^{\text{disp}}) \quad (\text{hidden: partner type and disposition}) \quad (1)$$

$$o_k = (o_k^{\text{act}}, o_k^{\text{pay}}) \quad (\text{observed: partner response and payoff}) \quad (2)$$

$$a \in \mathcal{A} \quad (\text{cooperate/defect or investment level}) \quad (3)$$

The model $\mathcal{M}_k = \{A_k^{\text{act}}, A_k^{\text{pay}}, B_k, C_k, D_k, E_k\}$ encodes likelihoods ($A_k^{\text{act}}, A_k^{\text{pay}}$), transition dynamics (B_k), prior beliefs (D_k), preferred observations (C_k), and policy priors (E_k). Payoff enters as an observation modality rather than an external reward cache, so the agent’s preferences are part of the same inference problem as its partner beliefs. State and policy inference use the official `pymdp` discrete active-inference implementation [9, 3].

2.3 Partner-local affective precision

Hesp et al. [10] treat the policy precision parameter γ as an inferred quantity rather than a fixed hyperparameter. The agent maintains a higher-order belief β over its expected action precision, representing subjective model fitness—how well the generative model is supporting confident action. An affective charge signal ϕ drives updates to β : it rises when the model performs better than a prior baseline and falls when it performs worse, making it a derivative of model fitness that responds to changes in predictive performance rather than to absolute reward level.

We extend this to the per-partner case. After interacting with partner k , the agent computes a bounded prediction-error proxy on the partner’s response:

$$\epsilon_k = 1 - P(o_k^{\text{act}} = \hat{o}_k^{\text{act}} \mid q(s_k^{\text{type}}, s_k^{\text{disp}})). \quad (4)$$

This is computed from partner *action* rather than payoff because action is the most direct test of type-by-disposition model fit: a reliably defecting partner produces no action surprise even though the payoff is poor. The proxy is bounded in $[0, 1]$ rather than log-surprisal, keeping affective charge in an interpretable range.

Affective charge is:

$$\phi_k = \alpha(\sigma_0^2 - \epsilon_k^2), \quad (5)$$

where σ_0^2 is a prior fitness baseline and α is a gain parameter. Positive ϕ_k means partner k ’s model predicted better than the baseline; negative ϕ_k means it predicted worse. A categorical posterior $q(\beta_k)$ over discrete levels $\{0.5, 0.67, 1.0, 1.5, 2.0\}$ is updated each round by ϕ_k . Higher β_k represents lower expected model fitness—the agent has accumulated evidence that its model of this partner is unreliable. The partner-specific policy precision is:

$$\gamma_k = \gamma_{\text{base}} / \mathbb{E}_{q(\beta_k)}[\beta_k]. \quad (6)$$

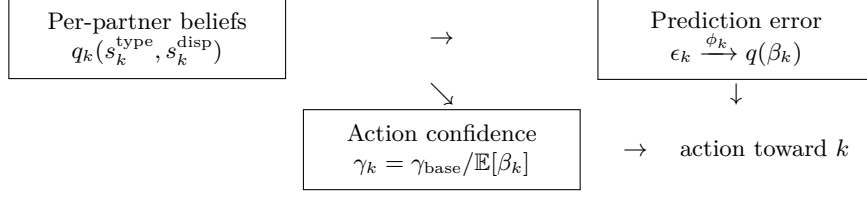


Fig. 1: Partner-local affective precision. Each partner has an independent belief state and precision estimate. Prediction error on partner action updates β_k , which sets how sharply the agent commits to policies involving that partner.

A well-validated partner model drives $\mathbb{E}[\beta_k]$ toward low values, raising γ_k and sharpening policy selection for that partner. An unreliable partner model keeps $\mathbb{E}[\beta_k]$ high, lowering γ_k and leaving policy selection diffuse. The affective precision channel thus translates accumulated evidence about predictive reliability into action confidence—independently for each social relationship.

The β_k posterior is maintained as an external auxiliary tracker rather than a proper hidden state in the generative model, which simplifies implementation at the cost of losing the agent’s ability to form prior expectations over its own future affective states (discussed in Section 4).

Tracked-only condition. To establish that any behavioral effect of the affective precision channel runs specifically through action confidence—rather than through indirect effects on belief quality—we include a tracked-only variant. This variant retains the β_k update in full but fixes $\gamma_k = \gamma_{\text{base}}$ for all partners at all times, severing the link from the precision estimate to the softmax temperature. The β_k posterior evolves normally and is available for analysis, but it never changes how policies are selected. If tracked-only behavior matches the no-affect baseline while full precision modulation differs from both, the behavioral effect runs through the γ_k pathway. If tracked-only matches full precision modulation, the effect is mediated by belief quality, not action sharpening.

3 Results

We report five sets of analyses. Section 3.1 establishes what the affective precision channel actually tracks. Section 3.2 shows that its behavioral effect depends on how open the policy space is, and that the effect runs through action sharpening rather than better partner beliefs. Section 3.3 examines partner selection. Sections 3.4 and 3.5 characterize the mechanism’s volatility limit. Section 3.6 reports precision-dynamics phenotypes as a parameter-sensitivity check.

3.1 The affective precision channel tracks partner predictability, not partner value

The central question for the affective precision channel is whether its running estimate of model fitness actually reflects predictive reliability or whether it is a proxy for experienced reward. A predictable exploiter is low value but easy to model; a valuable but erratic partner is worth engaging but hard to predict. If the channel tracks value, these two situations should produce similar precision levels. If it tracks model fitness, they should diverge.

We evaluate this in the binary game across 30 seeds and 200 rounds per seed. The affective precision channel (γ_k -derived signal) correlates more strongly with partner predictability than with payoff: $|\text{corr}(\text{precision}, \text{surprise})| = 0.701$ versus $|\text{corr}(\text{precision}, \text{payoff})| = 0.419$ across partner-seed units. The seed-level gap between these two correlations is $+0.096$, bootstrap CI $[0.027, 0.164]$. The channel is carrying partner-model reliability information, not reward history.

The binary regime also shows that having this reliability information does not automatically improve payoff: precision modulation yields 534.6 versus 542.1 for no-affect, mean difference -7.5 , CI $[-27.9, 13.8]$. This is the expected pattern. In the binary game the expected free energy gap between cooperating and defecting is large enough that the policy posterior is already near-deterministic; sharpening or softening a near-argmax changes almost nothing. The channel is informative about reliability but has nowhere to act. Section 3.2 shows what happens when it does.

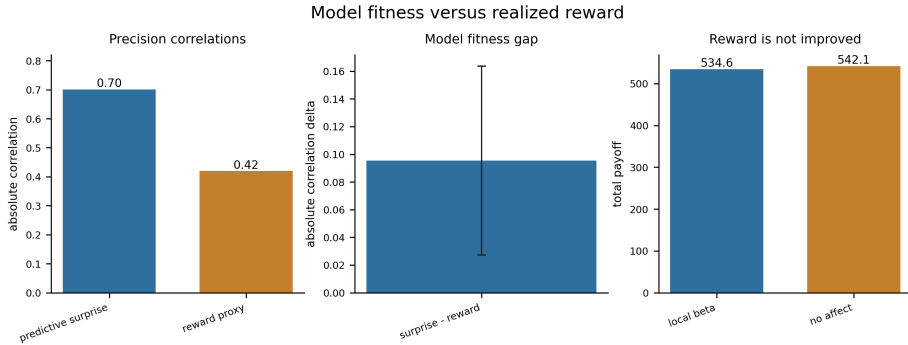


Fig. 2: The affective precision channel tracks partner predictability more strongly than realized reward. In the binary game the channel has no payoff advantage, consistent with a near-saturated policy space where precision cannot influence action.

3.2 Behavioral effects depend on policy entropy; action sharpening accounts for the benefit

Whether the affective precision channel can improve behavior depends on how much room the policy space gives it to act. We compare five seeds per variant, 200 rounds, across both regimes. In the binary game: no payoff advantage, no entropy difference (Section 3.1). In the graded investment game: precision modulation reaches total payoff 1884.6 versus 1859.4 for the no-affect baseline (+25.2), with mean policy entropy 7.94 versus 8.80 (−0.86). The improvement emerges precisely in the regime where policy entropy is high enough for action confidence to matter.

The tracked-only lesion establishes that this benefit comes from action sharpening, not from better partner beliefs. In the graded regime, the tracked-only agent—which updates β_k normally but never uses it to set γ_k —has joint type-disposition belief accuracy 0.339 and stance accuracy 0.855, slightly better than the precision-modulated agent’s 0.336 and 0.816. Its beliefs are at least as good, yet its payoff and entropy track the no-affect baseline rather than the precision-modulated agent. The behavioral gain is attributable specifically to the $\beta_k \rightarrow \gamma_k$ pathway. The affective precision channel augments social inference not by improving partner-model accuracy—something like better ToM—but by regulating how confidently those models are acted upon.

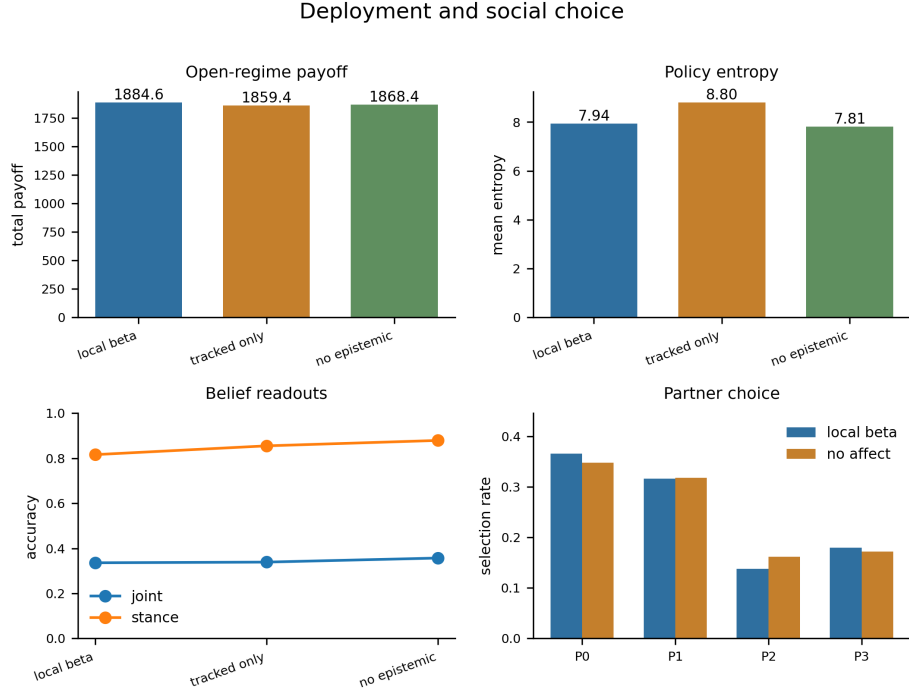


Fig. 3: Behavioral effects are regime-dependent and traceable to action sharpening. Precision modulation improves payoff and reduces policy entropy in the decision-open graded regime only. The tracked-only lesion matches the no-affect baseline despite maintaining comparable partner beliefs, identifying action confidence—not belief quality—as the locus of the effect.

3.3 Partner selection reorganizes around reliability

In partner-choice regimes the agent’s action-confidence estimates shape not just how it behaves toward a chosen partner, but which partners it chooses to engage in the first place. Aggregate payoff is a blunt measure here; subtle shifts in selection patterns can reflect meaningful reorientation before payoff has had time to accumulate.

Across five seeds and 200 rounds, aggregate payoff is nearly flat: 393.6 versus 393.2. But policy entropy under precision modulation is 3.99 versus 4.83 without it, and selection rates shift in the expected direction: the most predictable partner draws more engagement ($0.348 \rightarrow 0.366$), a less predictable partner draws less ($0.162 \rightarrow 0.138$). The affective precision channel is reorganizing social engagement around partner-model reliability—concentrating interaction where the agent can act confidently—at a scale visible in selection behavior before it registers in payoff.

3.4 Abrupt betrayal is the mechanism’s volatility limit

All conditions above involve stationary partners. Betrayal scenarios test what happens when a partner’s disposition changes mid-episode: the agent enters the post-switch period with a model that has been accumulating positive affective charge under the old behavioral regime.

The problem is precisely that the affective precision channel has been doing its job. The agent has built up action confidence toward the betraying partner based on a model that no longer applies. When the partner changes, the β_k posterior still reflects the pre-switch reliability record while the lower-level partner-state beliefs—which need repeated evidence to reorganize—are only beginning to update. The channel sharpens action based on accumulated confidence in a stale model.

Across 30 seeds and 120 rounds this is what we observe. Precision modulation yields lower whole-run payoff than the no-affect and tracked-only baselines: 1136.1 versus 1172.1, mean difference -36.1 , bootstrap CI $[-63.7, -10.9]$, $p = 0.017$. Policy entropy is lower under precision modulation (8.38 vs. 8.74), confirming the channel is active and influencing action. Post-switch, the agent returns to the switched partner less often (4.4 vs. 6.1 reencounters; selection rate 0.049 vs. 0.067) but with no calibration improvement: payoff per reencounter 8.76 vs. 8.91, wrong-type rate 0.237 vs. 0.165. The agent acted decisively; its inference did not justify that decisiveness.

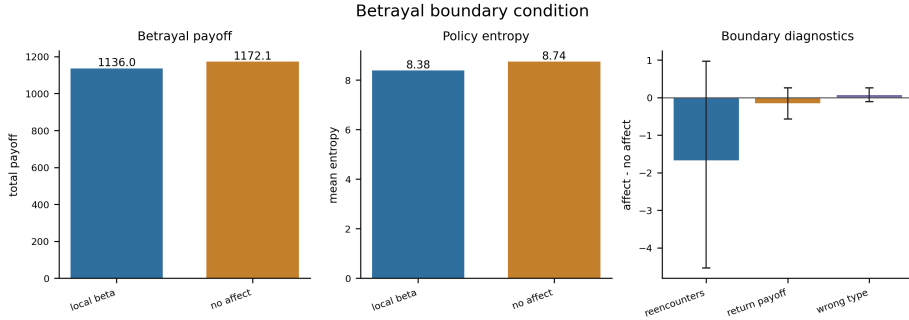


Fig.4: Abrupt betrayal is the volatility limit. Precision modulation reduces whole-run payoff, lowers policy entropy, and changes reallocation without improving calibration. The agent acts more confidently on a partner model that has become stale.

3.5 The limit is the speed of change, not the amount of precision

The betrayal failure might suggest the problem is too much precision gain—a β_k update that is too fast. The sensitivity analysis rules this out. A combined-caution variant that raises β globally to reduce precision gain produces worse

payoff, not better: 1115.0 versus 1172.1 for no-affect ($p = 0.003$, entropy 9.27). Reducing precision uniformly removes the useful action confidence in the stable portions of the episode without addressing the timing problem in the betrayal portion.

Gradual betrayal confirms that the limit is the speed of change. When the partner’s disposition shifts slowly, the β_k posterior tracks the changing prediction record before a large confidence surplus can accumulate. Default precision modulation is neutral under gradual change: payoff 1147.6 versus 1148.9 for no-affect, $p = 0.906$. Combined caution again hurts (1118.3, $p = 0.005$). The affective precision channel is adaptive when partner behavior is stable and neutral when it changes slowly; it becomes maladaptive specifically when behavioral change is faster than partner beliefs can track.

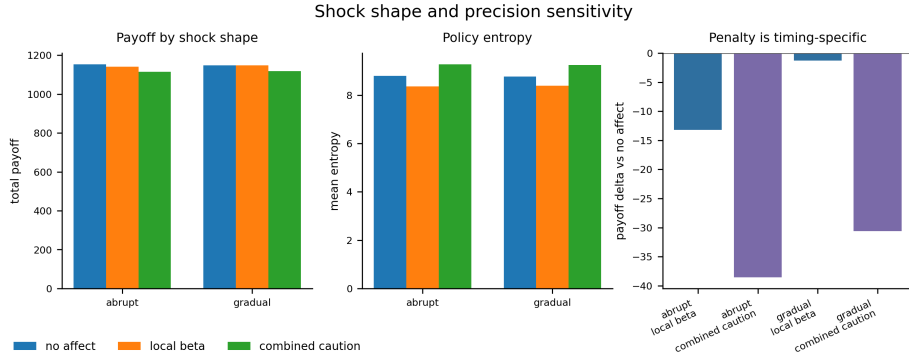


Fig. 5: Shock-shape sensitivity. Default precision modulation is neutral under gradual betrayal and harmful only under abrupt betrayal. Generic caution reduces payoff in both conditions. The mechanism’s limit is the timescale of behavioral change, not the magnitude of precision.

3.6 Precision-dynamics phenotypes

As a compact parameter check we verify that the affective precision channel responds to parameterization in an interpretable way. In the open graded regime a low-gain variant (small α) produces narrow β_k movement (mean range 0.149 vs. 0.950 for default); high-gain and cautious-prior variants produce wider movement (1.318 and 1.435). The same ordering holds in betrayal (0.132, 1.114, 1.091). Payoff does not rank variants monotonically by precision-dynamics range, consistent with the regime-dependence result: the affective channel’s behavioral consequence depends on the task context, not just on how much it moves.

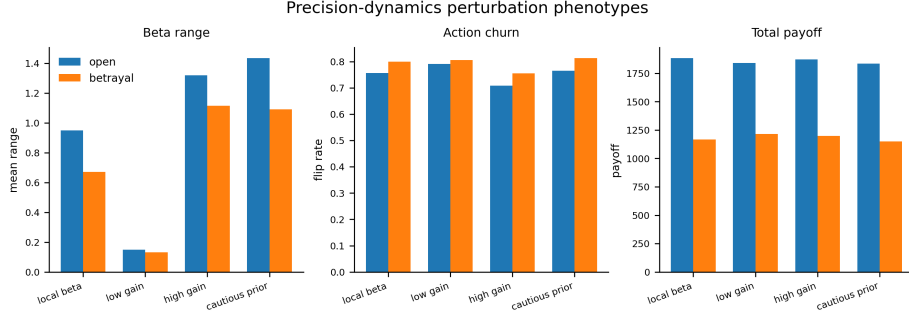


Fig. 6: Precision-dynamics phenotypes. Low-gain, high-gain, and cautious-prior variants separate clearly in β_k movement range. Behavioral consequences depend on the task regime rather than on the magnitude of precision dynamics alone.

4 Discussion

What the channel does. The core finding is a dissociation between two components of social action: what the agent infers about a partner, and how confidently those inferences guide behavior. The affective precision channel carries partner-model reliability information, not reward history. The tracked-only lesion shows that its behavioral effect is not explained by better partner beliefs—the lesion preserves, or slightly improves, belief accuracy while producing no behavioral gain. The effect is at the action-selection stage: the same partner-state beliefs, when coupled to a sharpened policy posterior, produce different choices.

This supports the framing of affect as an augmentation layer over social inference rather than as a reward signal or a belief-accuracy mechanism. An agent with good ToM-like partner models can still be systematically under- or over-confident about acting on them; the affective precision channel calibrates this confidence independently of model content. The tracked-only lesion makes this experimentally concrete: it shows that “better use of social beliefs” and “more accurate social beliefs” are separable outcomes, and that the affective channel contributes to the former.

Metacognitive, not mentalistic. The mechanism is worth distinguishing from theory-of-mind inference. ToM approaches model the agent as maintaining representations of what partners believe, know, or intend—a second-order model of another agent’s mental states [7, 14]. The affective precision channel does not do this. It tracks confidence in the focal agent’s own model of a partner’s *behavior*, not in representations of that partner’s mental states. This makes it metacognitive rather than mentalistic: the signal is about how well the agent’s own inference is performing, not about what the partner is thinking.

The two can complement each other. An agent with sophisticated ToM inference could still be poorly calibrated about when to act confidently on those

inferences—particularly if the partner is reliable in some contexts and unreliable in others. Partner-local affective precision addresses exactly this: it allows action confidence to track the recent track record of each partner model, independently of the model’s representational content. A natural extension is to combine ToM-style belief inference [15] with partner-local affective precision, so that the agent both models partner intentions and maintains a running estimate of how much those models currently justify decisive action.

The betrayal failure is theoretically informative. The abrupt-betrayal result is the strongest evidence that the mechanism is doing something real. If the affective precision channel were behaviorally inert, all betrayal variants would look the same. Instead, precision modulation lowers entropy, changes reallocation, and makes the agent more decisive—but around a stale model. This is the timing mismatch: the affective charge signal ϕ_k responds quickly to each interaction’s prediction error, driving rapid shifts in $\mathbb{E}[\beta_k]$ and therefore in γ_k , while partner-state beliefs—which require cumulative evidence across many rounds to reorganize—move slowly. After an abrupt switch there is a window during which action confidence reflects the pre-switch reliability record while lower-level inference is still catching up.

The gradual betrayal result confirms this interpretation directly: when behavioral change is slow enough that partner beliefs can track the shift before large confidence surplus accumulates, default precision modulation is neutral. The problem is not how much the affective channel responds, but how fast the social world changes relative to how fast beliefs reorganize.

The implication is that an effective affective precision channel in volatile environments would need a way to modulate the weight it places on recent charge signals based on how much the partner’s behavior appears to be changing. This is structurally similar to hierarchical volatility tracking in belief-updating models [12, 1], but applied to action confidence rather than to learning rates. Whether these two levels of volatility sensitivity—for beliefs and for action precision—operate independently or are coupled in human social cognition is an open empirical question that this work motivates.

Limitations. Three design features bound the current claims. First, β_k is an external auxiliary tracker; in the full Hesp et al. hierarchy it is a proper inferred hidden state, allowing the agent to have prior expectations about its own future affective states and to plan around anticipated changes in partner reliability. Making β_k a variational variable would enable richer prospective regulation of action confidence. Second, partners are scripted rather than themselves being active-inference agents; the mechanism is tested in isolation from the reciprocal dynamics that characterize real social interaction. Third, there is no global- β ablation yet—a condition that shares one affective precision estimate across all partners while preserving the same $\beta \rightarrow \gamma$ mapping. Without this, it remains possible that any precision modulation, not specifically partner-local precision, would produce similar benefits in the stable regimes.

Future directions. The global- β ablation is the most important near-term experiment. The predicted outcome is that a shared precision estimate would either over-suppress action confidence toward reliably predictable partners or under-suppress it toward uncertain ones, depending on the mix—but this should be tested. A second priority is a controlled shock-gradient study that varies betrayal abruptness parametrically, targeting the timescale boundary at which the mechanism transitions from neutral to harmful and testing the timing-mismatch account directly.

More broadly, the framework invites extension toward richer social environments. The trust game has no communication, reputation signaling, or coalition structure; all of these would introduce new sources of reliability evidence that the affective channel could in principle track. Testing with human participants would allow partner-local β dynamics to serve as a computational model of individual differences in social confidence calibration—with faster, higher-gain channels predicted to show stronger post-cue reallocation but greater sensitivity to abrupt social change. Finally, combining this mechanism with ToM-style belief inference [14] would allow future work to study whether metacognitive action-confidence signals and mentalistic partner-state models interact during social decision-making, and how they trade off under resource or time constraints.

5 Conclusion

Predicting what a partner will do and deciding how confidently to act on that prediction are separable problems. This paper shows that affect, formalized as a partner-specific estimate of social model fitness, can address the second problem. In repeated trust-game simulations, partner-local affective precision tracks partner predictability rather than partner value, improves performance through action sharpening in decision-open regimes, and reorganizes partner engagement around reliability before aggregate payoff separates. When partners change abruptly, the same mechanism sharpens action around a model that has not yet caught up—a failure that reveals the timing structure of the mechanism rather than a flaw in its steady-state design. Together, these results frame affect as an augmentation layer for social multi-agent inference: it helps agents decide when and how confidently to act on their social models, with a cost that is conditional on the volatility of the social world.

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